A decision-support framework for stock index forecasting and trading: Genetic-optimized VMD with convolutional LSTM and walk-forward validation

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Abstract

Under nonlinear and non-stationary dynamic environments, accurate stock index forecasting is essential for risk management and quantitative decision-making.This study investigates a stock investment decision support system comprising a hybrid neural network model and a trading decision application.In the hybrid neural network modeling component, a hybrid forecasting model termed GA-VMD-C-LSTM is proposed. The model employs a genetic algorithm (GA) to determine the optimal parameters for variational mode decomposition (VMD) and integrates long short-term memory (LSTM) networks with convolutional layers to capture complex temporal patterns in the data. Subsequently, a grid search is conducted to identify the optimal parameter combinations for the baseline GA-VMD-LSTM model, and comparative experiments with varying numbers of convolutional layers are performed to determine the optimal number of convolutional layers. Furthermore, the GA-VMD-C-LSTM model is compared with several benchmark models. The empirical results confirm the effectiveness of the convolutional layers and indicate that the proposed model outperforms the benchmark models.Compared with the best-performing benchmark model, the proposed GA-VMD-C-LSTM achieves performance improvements, with the mean absolute error (MAE) reduced by 46.72%, the root mean squared error (RMSE)

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reduced by 44.63%, the mean absolute percentage error (MAPE) reduced by 46.31%, and the coefficient of determination R^2 increased by 0.03%. In the trading decision application component, a threshold-based timing strategy considering transaction costs is employed to evaluate trading utility, thereby assessing the practical applicability of the proposed model. Overall, this study provides a technical approach for investment decision-making and risk management in financial systems.

Keywords:
Stock index forecasting, Decision support system, Variational mode decomposition, Genetic algorithm, Convolutional LSTM, Multi-scale time series, Algorithmic trading

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1. Introduction

The stock market plays a crucial role in the global economy by providing substantial returns to investors while exposing them to significant risks. Accurate prediction of stock price movements helps market prices reflect their intrinsic value (Zhu et al., 2024; Yu et al., 2025), which is of considerable practical importance for investors, financial institutions, and regulatory authorities.

Stock prices are influenced by various factors, including macroeconomic fundamentals, monetary and liquidity conditions, investor sentiment, and international commodity prices. From a macroeconomic perspective, variables such as interest rates, inflation, and exchange rates exert significant long-term constraints on stock price movements (Keswani et al., 2024). Additionally, stock prices are closely associated with indicators such as Gross Domestic Product (GDP) and inflation (Humpe et al., 2025). Beyond traditional macroeconomic factors, expansions in the money supply can directly increase demand for stocks, driving up prices and leading to a systematic divergence between stock prices and economic fundamentals (Hirota, 2023). Investor sentiment not only affects price volatility but also interacts with stock price jump risks through a dual-layer network structure, exhibiting stronger spillover effects within short-term risk networks (Gao & Zhao, 2023). Oil prices also play an important role in stock price dynamics, as oil price trends are negatively associated with future stock returns, implying that rising oil price trends tend to predict lower future excess returns (Cao et al., 2022).

Stock prices are typically characterized by non-stationarity and nonlinearity, which makes stock price forecasting a challenging task. Early studies on stock price prediction primarily relied on traditional time series models (Idrees et al., 2019). In recent years, with advances in computing technology, machine learning methods have been increasingly applied to stock price forecasting (Chen & Hao, 2017). Deep learning, a subfield of machine learning, is based on deep neural networks and eliminates the need for manual feature engineering, achieving higher predictive accuracy than conventional machine learning models (Petropoulos et al., 2022). As neural networks constitute a core component of deep learning and represent a research frontier across various scientific disciplines (Dargan et al., 2020), this study employs neural network–based models to forecast stock index prices.

In recent years, forecasting models and decision support systems based on intelligent algorithms have gained increasing attention in the field of financial investment. A swing trading decision support system utilizing long short-term memory (LSTM) networks provides traders with investment analysis by integrating multiple technical indicators (Banik et al., 2022). To address the limited interpretability associated with black-box models in financial forecasting, an interpretable decision support system based on multi-stage feature selection and classification and regression trees has been proposed. This system enhances model transparency and decision credibility while maintaining predictive performance (Dolatsara et al., 2022). Furthermore, integrated decision support frameworks that consider the entire investment process combine price forecasting, stock selection, and portfolio optimization. These frameworks enable multi-stage coordinated decision-making through intelligent optimization algorithms and achieve investment performance comparable or superior to benchmark strategies across diverse market environments (Solares et al., 2022). Consequently, in this study, forecasting models are integrated with decision support mechanisms.

Although previous studies have proposed various effective models for stock price forecasting, there remains room for improvement. First, most existing models determine the parameters of Variational Mode Decomposition (VMD) based on experience or trial-and-error. Such parameter settings do not guarantee optimal decomposition performance, which may, in turn, reduce forecasting accuracy. Second, many studies employing frequency decomposition techniques forecast each Intrinsic Mode Function (IMF) and the residual separately before reconstructing the final prediction, often without considering the correlations among the decomposed components. Third, most existing forecasting models rely on standard LSTM architectures. Although LSTM networks can capture temporal dependencies, they are less effective at extracting local features from time series, limiting their ability to accurately predict stock prices. Fourth, existing machine learning-based studies often lack a thorough investigation into the practical applicability of the proposed models. Therefore, to address these limitations, a decision support system is proposed, consisting of a hybrid neural network forecasting model (GA-VMD-C-LSTM) and a trading decision application built upon this model. Specifically, the proposed forecasting approach involves employing a genetic algorithm (GA) to optimize the parameters of variational mode decomposition (VMD) and incorporating convolutional layers into the LSTM network to extract local information from stock price time series.

The main contributions of this study are summarized as follows:(1) A Genetic Algorithm (GA) is employed to search for the optimal VMD parameters, specifically K, α, and τ, to improve the quality of signal decomposition. Niu et al. (2020) proposed a VMD–LSTM model for stock price forecasting in which parameters were determined empirically. However, such settings may negatively affect decomposition performance. As an improvement, Zhang et al. (2023) and Yu et al. (2025) applied GA to optimize K and α but omitted the parameter τ. This study jointly optimizes all three parameters (K, α, and τ) to further enhance decomposition performance.(2) The optimized VMD algorithm generates decomposed components that are fed into the neural network simultaneously, rather than being predicted independently. Yujun et al. (2021) integrated VMD with LSTM by forecasting each component separately. In contrast, this study accounts for the correlations among VMD-derived components by inputting all components into the network jointly.(3) A convolutional layer is introduced into the GA-VMD-LSTM framework, and the impact of the number of convolutional layers on forecasting performance is systematically investigated. While the GA-VMD-LSTM model proposed by Huang et al. (2021) does not explicitly capture local features in the time series, the architecture in this study incorporates a convolutional layer to address this limitation.(4) A market timing strategy is implemented to evaluate the returns of the proposed model in a simulated trading environment, thereby validating its practical applicability and robustness.

The remainder of this paper is organized as follows. Section 2 reviews related work on stock price forecasting, covering traditional time series models and deep learning approaches. Section 3 describes the methodology employed in this study, including the proposed hybrid architecture. Section 4 presents the experimental design and empirical results, providing a comparative analysis against benchmark models. Finally, Section 5 concludes the paper and discusses directions for future research.

2. Literature review

Early studies on stock price forecasting primarily focused on traditional econometric models. Initial econometric approaches were mainly designed for stationary time series. To address the pervasive trends and non-stationarity observed in real-world financial data, Box & Jenkins (1970) proposed the Autoregressive Integrated Moving Average (ARIMA) model, which transforms non-stationary series into stationary ones through differencing prior to modeling. Subsequently, Bollerslev (1986) developed the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, enabling volatility to be modeled as a dynamic process and successfully capturing the volatility clustering inherent in financial time series.

Building on multifractal volatility (MFV) measures, Wei & Wang (2008) employed an ARFIMA model to forecast the volatility of high-frequency data from the Shanghai Stock Exchange (SSE) Composite Index. Their results demonstrated that ARFIMA outperformed traditional models, such as realized volatility (RV), GARCH, and stochastic volatility (SV), in one-day-ahead volatility forecasting. However, Sáenz et al. (2023) showed that when ARIMA models are applied to stock price prediction, their performance is constrained by the pronounced nonlinear dependence and complex dynamic structures inherent in financial markets.

Some studies have employed single-model architectures for stock price forecasting. Ozcalici & Bumin (2022) utilized artificial neural networks (ANNs) to construct a stock screening framework, where performance indicators from the training set were used to predict the performance of individual stocks in the testing set, thereby selecting portfolios expected to generate higher returns. Empirical results showed that ANN-selected portfolios achieved higher returns than the market benchmark. Ma & Yan (2022) applied convolutional neural networks (CNNs) to stock market forecasting and reported an average prediction accuracy of approximately 70%, which outperformed traditional machine learning methods. With the advancement of deep learning, long short-term memory (LSTM) networks have increasingly been adopted for financial time series analysis. Ghosh et al. (2022)

employed LSTM models to predict price movements across multiple assets, demonstrating that LSTM outperformed random forests and support vector machines in capturing temporal dependencies and nonlinear relationships. Their experimental results further confirmed the generalization capability and robustness of LSTM in multifactor forecasting. Additionally, Firouzjaee & Khalilian (2024) applied LSTM models to forecast energy stock prices by incorporating macroeconomic variables, such as crude oil prices, gold prices, and the U.S. dollar index, and examined the impact of feature correlations on predictive performance and model interpretability.

To further enhance forecasting accuracy and robustness, some researchers have combined multiple neural network architectures. Jing et al. (2021) proposed a CNN–LSTM model integrated with investor sentiment analysis for stock price prediction, finding that the hybrid framework improved both stability and accuracy. Barua & Sharma (2022) developed a CNN–BiLSTM hybrid neural network that outperformed standalone CNN or LSTM models in stock index forecasting, achieving high-precision multi-step predictions. Kanwal et al. (2022) proposed a GPU-accelerated hybrid deep learning model, BiCuDNNLSTM–1D CNN, which integrates bidirectional CuDNNLSTM with one-dimensional convolutional neural networks. This model is capable of capturing global temporal dependencies while extracting local features within sliding time windows.

In recent years, with the rapid development of signal processing techniques, frequency decomposition algorithms have been increasingly combined with neural networks for stock price forecasting. Cao et al. (2019) integrated Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) and LSTM to handle nonlinear and non-stationary financial time series. Niu et al. (2020) combined the decomposition capability of Variational Mode Decomposition (VMD) with the temporal learning strength of LSTM to construct a VMD–LSTM model, demonstrating the effectiveness of decomposition–reconstruction frameworks for complex financial series. Yujun et al. (2021) also proposed a hybrid model integrating VMD and LSTM for stock price prediction. To address the strong nonlinearity and volatility of financial data, Rezaei et al. (2021) developed a CEEMD–CNN–LSTM model to forecast stock indices such as the S&P 500.Yao et al. (2023) proposed a hybrid forecasting framework based on Multivariate Empirical Mode Decomposition (MEMD) and Temporal Convolutional Networks (TCN); empirical results across multiple markets showed that this model outperformed various benchmarks in terms of forecasting accuracy, directional symmetry, and stability. Building upon this line of research, Zhu et al. (2024) proposed a CEEMDAN–S–C–LSTM hybrid model, which combines CEEMDAN with Savitzky–Golay (SG) filtering for noise reduction, a three-layer convolutional network for local feature extraction, and an LSTM layer for long-term dependencies. The results indicated that this model outperformed conventional LSTM and CEEMDAN–LSTM architectures. More recently, Li et al. (2025) developed a CEEMDAN–Informer–LSTM framework for forecasting the CSI 300 index. In this model, CEEMDAN decomposes the series into Intrinsic Mode Functions (IMFs) at different frequencies, after which the Informer and LSTM models are used to forecast high-frequency and low-frequency components, respectively, to capture both short-term fluctuations and long-term trends.

Some researchers have utilized various optimization algorithms to identify the optimal parameters for Variational Mode Decomposition (VMD). VMD is characterized by its adaptive and non-recursive properties, which effectively alleviate mode mixing (Dragomiretskiy & Zosso, 2013). However, the selection of VMD parameters is critical to decomposition quality and subsequent forecasting performance (Lin et al., 2022). Yu et al. (2023) applied the Grey Wolf Optimizer (GWO) to search for optimal VMD parameters and proposed a GVMD–Q–DBN–LSTM–GRU model to forecast the volatility of the Shanghai Composite, S&P 500, and FTSE indices. Wang et al. (2025) optimized VMD parameters using Particle Swarm Optimization (PSO), employed TimeGAN to impute missing values, and captured linear structures with ARIMA. The outputs of CNN–BiLSTM and XGBoost were then weighted, yielding results that outperformed fifteen benchmark models. Huang et al. (2021) proposed a GA-VMD-LSTM (GVL) model for financial forecasting, in which a Genetic Algorithm (GA) was used to optimize VMD parameters.Zhang et al. (2023) developed a VMD–CNN–BiLSTM–MLP framework, where GA optimized the VMD parameters, while Multilayer Perceptrons (MLP) and CNN–BiLSTM modeled high-frequency noise IMFs and low-frequency trend IMFs, respectively. This approach achieved stable forecasting for EU Allowance (EUA) carbon futures across two policy regimes. Yu et al. (2025) proposed a hybrid GA-VMD-TCN model that integrates GA-optimized VMD with Temporal Convolutional Networks (TCN) for multi-step forecasting of the Shanghai Composite and S&P 500 indices. This framework demonstrated effectiveness in parameter adaptivity, nonlinear feature extraction, and multiscale dynamic modeling.

Table 1 summarizes representative models used for stock price prediction. Overall, research has shifted from traditional statistical models toward deep learning-based and hybrid architectures. Among these, LSTM-based frameworks remain prominent, reflecting their potential in stock price prediction. Recently, hybrid models combining frequency decomposition with complex neural networks have gained attention, with an increasing number of studies incorporating optimization algorithms. Building upon this prior research, this study proposes a hybrid forecasting model, the details of which are presented in the following section.

Table 1: Models for predicting stock prices.

Article	Models	Types
Box & Jenkins (1970),Sáenz et al. (2023)	ARIMA	Conventional Forecasting Models
Bollerslev (1986)	GARCH	Conventional Forecasting Models
Wei & Wang (2008)	ARFIMA	Conventional Forecasting Models
Ozcalici & Bumin (2022)	ANN	Deep Learning Predictive Models
Ma & Yan (2022)	CNN	Deep Learning Predictive Models
Ghosh et al. (2022),Firouzjaee & Khalilian (2024)	LSTM	Deep Learning Predictive Models
Jing et al. (2021)	CNN-LSTM	Deep Learning Predictive Models
Barua & Sharma (2022)	CNN-BiLSTM	Deep Learning Predictive Models
Kanwal et al. (2022)	BiCuDNNLSTM-1dCNN	Deep Learning Predictive Models
Cao et al. (2019)	CEEMDAN-LSTM	Hybrid Predictive Models
Niu et al. (2020),Yujun et al. (2021)	VMD-LSTM	Hybrid Predictive Models
Rezaei et al. (2021)	CEEMD-CNN-LSTM	Hybrid Predictive Models
Yao et al. (2023)	MEMD-TCN	Hybrid Predictive Models
Zhu et al. (2024)	CEEMDAN-S-C-LSTM	Hybrid Predictive Models
Li et al. (2025)	CEEMDAN-Informer-LSTM	Hybrid Predictive Models
Yu et al. (2023)	GVMD-Q-DBN-LSTM-GRU	Hybrid Predictive Models
Yu et al. (2025)	GA-VMD-TCN	Hybrid Predictive Models

Conventional Forecasting Models: conventional techniques for predicting stock prices.
Deep Learning Predictive Models: deep learning stock price prediction methods.
Hybrid Predictive Models: hybrid models combine deep learning and other stock price prediction methods.

3. Methodology

The primary methodologies employed in this study include genetic algorithms, frequency decomposition techniques, and neural network models. First, the genetic algorithm and the frequency decomposition method are introduced, as they constitute the preprocessing stage of the proposed hybrid framework. Subsequently, the LSTM model is presented, and the mechanism by which information is processed within the network is explained. Finally, the overall architecture of the proposed hybrid model is described.

3.1. Genetic algorithm (GA)

The genetic algorithm (GA), originally proposed by Holland (1975), is a global optimization method inspired by the mechanism of natural selection. By iteratively applying selection, crossover, and mutation operators, GA enables a population of candidate solutions to evolve toward an optimal solution. Since GA does not require the objective function to be differentiable or convex, it is effective in solving nonlinear, high-dimensional, and non-convex optimization problems. Consequently, GA has been widely applied to hyperparameter optimization, combinatorial optimization, and time series modeling (Golberg, 1989; Yu et al., 2025). Its primary strengths include global search capability, the ability to avoid local optima, and robustness.

Mathematically, let the variables to be optimized be defined as

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n) \tag{1}$$

GA first initializes a population randomly and evaluates the quality of candidate solutions using a fitness function. For minimization problems, the fitness function can be constructed as

$$\text{Fitness}(\boldsymbol{\theta}) = \frac{1}{f(\boldsymbol{\theta}) + \varepsilon} \tag{2}$$

where $f(\boldsymbol{\theta})$ denotes the objective function and ε is a small positive constant introduced to prevent division by zero. Subsequently, the selection operator preserves individuals with higher fitness, the crossover operator generates new parameter combinations, and the mutation operator introduces random perturbations to maintain population diversity. When the fitness converges or the maximum number of generations is reached, the GA terminates and outputs the optimal solution:

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \text{Fitness}(\boldsymbol{\theta}) \tag{3}$$

Through this evolutionary mechanism, GA can obtain high-quality solutions in complex search spaces, making it suitable for parameter optimization tasks that require global search.

3.2. Variational mode decomposition (VMD)

Variational Mode Decomposition (VMD) is an adaptive signal decomposition method based on a variational framework, designed to decompose a complex signal into a finite number of Intrinsic Mode Functions (IMFs) with limited bandwidth. By formulating a constrained variational optimization problem, VMD iteratively estimates the center frequency and the corresponding mode of each component in the frequency domain, ensuring the spectral content of each mode remains compact. This mechanism effectively alleviates mode mixing. Due to its fully non-recursive formulation, explicit parameterized control, and robustness to noise, VMD is particularly suitable for analyzing nonlinear and non-stationary financial time series (Dragomiretskiy & Zosso, 2013).

The specific procedure of VMD is described as follows.

VMD assumes that a given signal $f(t)$ can be represented as the sum of K band-limited intrinsic mode functions:

$$f(t) = \sum_{k=1}^{K} u_k(t), u_k(t) = A_k(t)\cos\big(\phi_k(t)\big) \tag{4}$$

where $A_k(t)$ denotes the slowly varying amplitude and $\phi_k(t)$ represents the instantaneous phase. VMD seeks to determine the set of modes $\{u_k, \omega_k\}$ by minimizing the sum of the bandwidths of all modes:

$$\min_{\{u_k\},\{\omega_k\}} \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2, \text{s.t. } \sum_{k=1}^{K} u_k(t) = f(t) \tag{5}$$

where the constraint ensures that the sum of all modes reconstructs the original signal.

By introducing the Lagrange multiplier $\lambda(t)$ and a quadratic penalty term with $\alpha > 0$, the augmented Lagrangian function is constructed as

$$\mathcal{L} = \sum_{k=1}^{K} \left\| \partial_t \big[\mathcal{H}\{u_k\}(t)\, e^{-j\omega_k t} \big] \right\|_2^2 + \left\langle \lambda,\ f - \sum_k u_k \right\rangle + \frac{\alpha}{2} \left\| f - \sum_k u_k \right\|_2^2 \tag{6}$$

where $\mathcal{H}$ denotes the Hilbert transform. The optimization problem is solved iteratively using the alternating direction method of multipliers (ADMM), with the following update rules:

$$\omega_k^{(n+1)} = \frac{\displaystyle\int_0^\infty \omega \left| \hat{u}_k^{(n+1)}(\omega) \right|^2 d\omega}{\displaystyle\int_0^\infty \left| \hat{u}_k^{(n+1)}(\omega) \right|^2 d\omega} \tag{7}$$

$$\hat{\lambda}^{(n+1)}(\omega) = \hat{\lambda}^{(n)}(\omega) + \tau \left[\hat{f}(\omega) - \sum_k \hat{u}_k^{(n+1)}(\omega) \right] \tag{8}$$

3.3. Long short-term memory

Recurrent Neural Networks (RNNs) have been widely applied in sequence modeling tasks, such as speech recognition, natural language processing, and financial time series forecasting, due to their ability to process sequential data. However, traditional RNNs often suffer from vanishing and exploding gradient problems during training, making it difficult to capture long-term dependencies. To address this issue, Hochreiter & Schmidhuber (1997) proposed the Long Short-Term Memory (LSTM) network. In an LSTM network, the input gate, forget gate, and output gate jointly regulate the update and transmission of information within the memory cell. The input gate determines which portions of the current input are written into the memory cell; the forget gate selectively retains or discards historical information; and the output gate controls the information flow from the memory cell to the hidden state. Through the coordinated operation of these three gating mechanisms, LSTM effectively mitigates the vanishing and exploding gradient problems (Hochreiter & Schmidhuber, 1997).

An LSTM network consists of multiple memory cells, each of which dynamically controls information flow through gating mechanisms. The structure of an LSTM cell is illustrated in Fig. 1. Let x_t denote the input sequence at time t, h_t the hidden state, c_t the cell state, and $\tilde{c}_t$ the candidate cell state. The computational process of LSTM is described as follows:

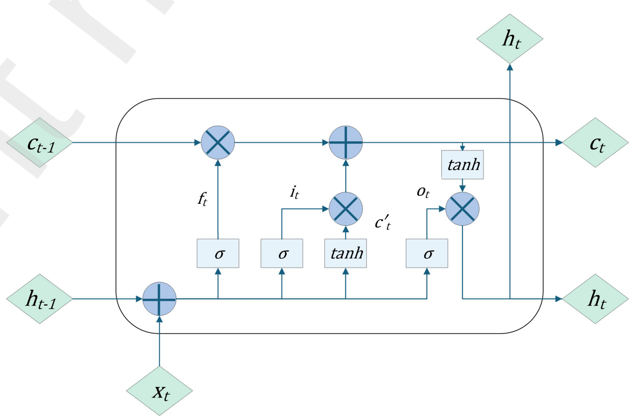

Figure 1: Schematic diagram of the LSTM architecture (Zhang et al., 2023).

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{9}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{10}$$

$$\tilde{c}t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{11}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}t \tag{12}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{13}$$

$$h_t = o_t \odot \tanh(c_t) \tag{14}$$

where σ denotes the sigmoid function; tanh represents the hyperbolic tangent function; $\odot$ indicates element-wise multiplication; f_t, i_t, and o_t correspond to the forget gate, input gate, and output gate, respectively; W_f, W_i, W_c, and W_o denote the weight matrices; and b_f, b_i, b_c, and b_o represent the bias vectors.

3.4. Proposed framework

This study proposes a decision support system for stock price forecasting and timing, consisting of a financial time series forecasting model that integrates multiscale decomposition with deep temporal modeling (GA-VMD-C-LSTM), and a trading decision application built upon this hybrid architecture. Regarding the hybrid forecasting model, the framework is designed to address the inherent non-stationarity and high-noise characteristics of price series, comprising a data preprocessing module and a neural network module.The core approach involves employing a genetic algorithm (GA) to determine the optimal values for the parameters K, α, and τ. These optimized values are then utilized as hyperparameters for Variational Mode Decomposition (VMD), which decomposes the complex price series into multiple intrinsic mode functions (IMFs) with distinct center frequencies. Subsequently, all decomposed modes are jointly fed into the C-LSTM network to generate the final forecasts. This model facilitates both local feature extraction and the capture of long-term dependencies, enabling a more comprehensive representation of the dynamic structure of financial markets. The overall framework of the proposed approach is illustrated in Fig. 2.

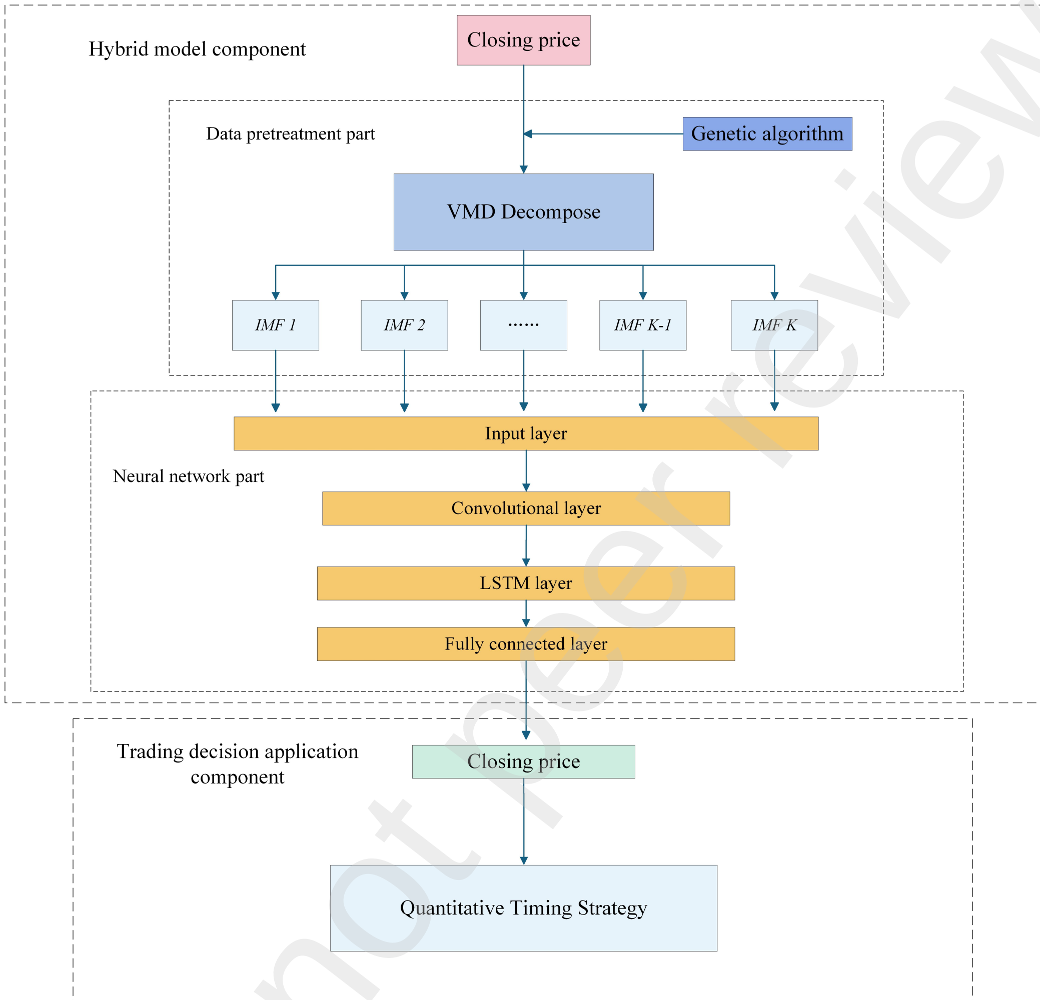

Figure 2: The architecture of the proposed prediction method.

In the preprocessing stage, GA and VMD are applied to the normalized original time series. Given the complexity and variability of stock indices, determining optimal parameters for VMD is challenging; an insufficient number of modes (K) may lead to mode mixing, whereas an excessively large K increases computational complexity (Yu et al., 2023; Yang et al., 2024). Furthermore, the penalty factor α helps mitigate Gaussian noise interference. While most studies utilizing GA for VMD optimization focus solely on K and α (Zhang et al., 2023; Yu et al., 2025), the update step τ is often neglected. However, an inappropriate τ can adversely affect the convergence and stability of the VMD process, thereby reducing forecasting accuracy. To enhance decomposition performance, this study incorporates the step size τ into the GA search space, performing a joint optimization of (K, α, τ).Specifically, Sample Entropy (SE) is employed as the fitness function for GA to optimize the key parameters (Richman & Moorman, 2000). In this process, GA treats each set of parameters as an "individual" and searches for the optimal combination through selection, crossover, and mutation operators. After evaluating each individual using the fitness function (see Eq. (15)), the highest-fitness individuals are preserved. Upon convergence of the GA iterations, the optimal K, α, and τ are obtained to configure the VMD for decomposing the original sequence. The optimization process is depicted in Fig. 3.

$$\mathrm{SampEn}(m, r, N) = -\ln\left(\frac{A^m}{B^m}\right) \tag{15}$$

Here, N denotes the length of the time series and m represents the embedding dimension ($m = 3$). B^m is the average probability that vectors of length m match within a threshold $r = 0.25 \times \mathrm{Std}$, while A^m is the average probability for vectors of length $m + 1$. The resulting Sample Entropy (SampEn) quantifies the complexity of the sequence, where higher values indicate greater irregularity and complexity.

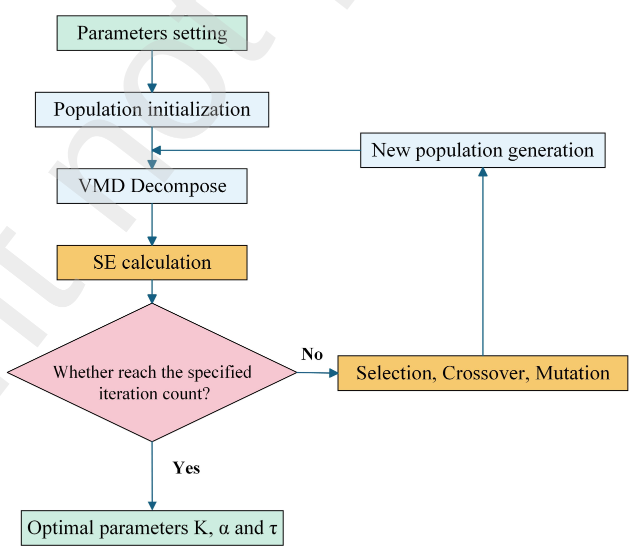

Figure 3: The flowchart for GA to find the optimal parameters K, α, and τ(Yu et al., 2025).

The neural network component consists of four types of layers: the input layer, convolutional layer, LSTM layer, and fully connected layer.

Regarding the input layer, Niu et al. (2020) and Yujun et al. (2021) fed the components obtained from VMD separately into subsequent LSTM networks and reconstructed the final prediction by aggregating the outputs. However, this strategy ignores the correlations among the decomposed components. Therefore, in this study, all components obtained from VMD are jointly fed into the network. The input layer introduces the decomposed data for further processing, and the number of neurons in this layer equals the number of decomposed components.

The convolutional layer is positioned after the input layer. Incorporating a convolutional layer enables the extraction of local information from stock price series, thereby improving forecasting accuracy (Zhu et al., 2024). The convolutional layer employed here differs from that in Rezaei et al. (2021) as the pooling layer is omitted. The convolution operation can be regarded as a weighted average of the input, where the weights constitute a filter (Hoseinzade & Haratizadeh, 2019). By assigning different weights to the input sequence, the filter extracts local features, such as price trend changes and local fluctuation patterns. The kernel size determines the range of local information extraction, with larger sizes corresponding to a wider temporal receptive field.

In this study, a one-dimensional (1D) convolutional layer is adopted, as illustrated in Fig. 4.

$$x_t^l = \sum_{i=0}^{k-1} w_{i+1} \cdot x_{t+i}^{l-1} + b \tag{16}$$

$$T' = T - k + 1 \tag{17}$$

The parameters of the 1D convolution are defined as follows: l denotes the l-th convolutional layer; x_t^{l-1} represents the input value; w_i denotes the convolutional weights; k is the kernel size, indicating the number of consecutive input points covered by each operation; b is the bias term; x_t^l represents the output at sliding position t, reflecting the matching degree between the input sequence and the kernel; T denotes the length of the input sequence; and T' represents the length of the output sequence. Since valid convolution is adopted without padding, the output length is shorter than the input length. After computing x_t^l, the filter slides forward to compute the next value x_{t+1}^l until the entire sequence is covered.

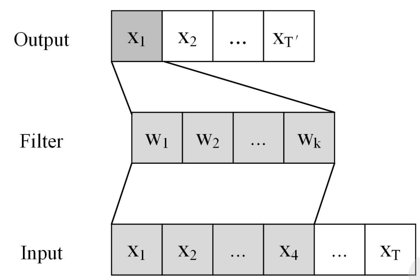

Figure 4: The schematic diagram of one-dimensional convolution operation(Zhu et al., 2024).

The LSTM layer receives the output from the convolutional layer and learns long-term dependencies through its gating mechanisms, producing a high-dimensional feature vector.

The fully connected layer then reduces this vector to a scalar, which corresponds to the final prediction, using the Rectified Linear Unit (ReLU) as the activation function.

The proposed hybrid model is denoted as GA-VMD-C-LSTM, where "C" represents the convolutional layer. Regarding parameter settings, following previous studies, the GA population size is set to 10, the crossover probability to 0.8, and the mutation probability to 0.025 (Yu et al., 2025). The convolutional kernel size is set to 3, with a padding length of 1, and the number of input channels equals the number of output channels (Zhu et al., 2024). Additionally, based on grid search results, the number of LSTM hidden units is set to 64, the time window length to 20, and the forecasting horizon to 1.

4. Experiments and results

This section identifies the optimal model parameters through grid search and investigates the impact of the number of convolutional layers on forecasting performance. Subsequently, the superiority of the proposed model is demonstrated through multi-model comparisons, and its practical applicability is validated via a market timing strategy experiment. Both the proposed and benchmark models are trained using the same dataset and then applied to out-of-sample data to predict closing prices. Performance is evaluated using several metrics. Before presenting the results, the experimental dataset and evaluation metrics are introduced.

4.1. Dataset

In this study, the CSI 300 Index is selected as the research sample, with the closing price serving as the input feature. The sample period spans from April 11, 2005, to August 22, 2025, comprising 4,952 daily observations. Descriptive statistics for the CSI 300 Index closing prices are reported in Table 2, and the original time series is illustrated in Fig. 5. The data are sourced from the CSMAR database.

To stabilize the training process and improve gradient update efficiency, the closing prices are normalized prior to modeling. The widely used Min–Max transformation is adopted, as defined follows:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{18}$$

where x denotes the original observation, $x_{\min}$ and $x_{\max}$ represent the minimum and maximum values of the variable over the sample period, respectively, and x' denotes the normalized value.

Following normalization, the dataset is divided into training and testing sets. Specifically, the first 70% of the samples are used for training, while the remaining 30% are reserved for testing. The trained model takes the testing data as input to generate forecasts. Each input sample consists of normalized features over 20 consecutive trading days; that is, the closing price of the subsequent trading day is predicted using information from the preceding 20 trading days.

Once the forecasting process is complete, an inverse normalization transformation (denormalization) is applied to map the predicted values back to the original price scale, facilitating direct comparison with observed prices. This transformation is expressed as:

$$\hat{x} = x'(x_{\max} - x_{\min}) + x_{\min} \tag{19}$$

where $\hat{x}$ denotes the denormalized prediction. Through this inverse transformation, the model outputs are restored to the original price level, enabling error measurement and performance evaluation against the actual values.

Table 2: Descriptive statistics of the closing prices of the CSI 300 Index.

Index	Sum	Mean	Standard deviation	Minimum	Q1 (25%)	Median	Q3 (75%)	Maximum
CSI 300	4952	3294.50	1043.77	818.03	2574.03	3396.46	3930.83	5877.20

Figure 5: The original sequence of the closing prices of the CSI 300 Index.

4.2. Evaluation metrics

To comprehensively assess the performance of the proposed model in regression forecasting tasks, four widely used evaluation metrics are employed: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2) (Zhu et al., 2024). Their mathematical definitions are provided below.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{20}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{21}$$

$$\mathrm{MAPE} = \frac{100}{n}\sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{22}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{23}$$

where y_i denotes the observed value, $\hat{y}_i$ represents the predicted value, and n is the total number of samples. Smaller values of MAE, RMSE, and MAPE indicate better forecasting performance, whereas a larger R^2 implies higher model explanatory power.

4.3. Model experimental results

A series of experiments are conducted to validate the effectiveness of the GA-VMD-C-LSTM hybrid model, including grid search for parameter selection, determination of the optimal number of convolutional layers, multi-model comparisons, and market timing strategy evaluations. Fig. 6 presents the decomposition results of the CSI 300 Index closing prices. As shown, the closing price series is decomposed into 13 Intrinsic Mode Functions (IMFs) using the VMD algorithm. The initial IMFs correspond to lower-frequency components with smoother variations, whereas the subsequent IMFs exhibit higher-frequency components with more pronounced fluctuations. The decomposed series provide a detailed representation of the frequency characteristics inherent in the original closing price. Fig. 7 illustrates the comparison between the forecasting results of the GA-VMD-C-LSTM hybrid model and the actual closing price series.

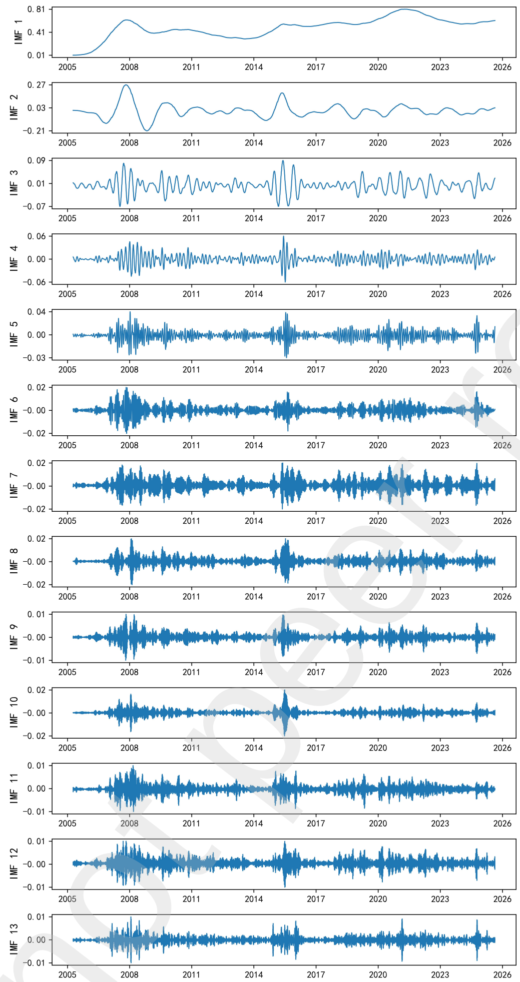

Figure 6: VMD decomposition result.

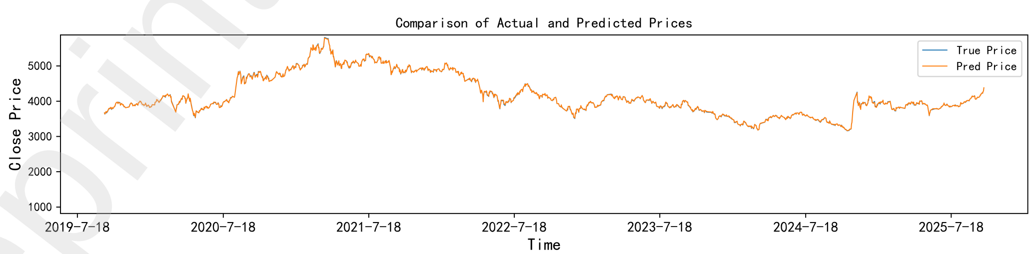

Figure 7: Comparison chart of actual price and predicted price.

In the first experiment, the GA-VMD-LSTM model is adopted as the baseline, and three key hyperparameters: seq_len, horizon, and hidden_size are optimized using a grid search strategy. Specifically, seq_len denotes the length of the input time window, with candidate values of 5, 20, and 60; horizon represents the forecasting horizon, taking values of 1, 5, and 10; and hidden_size refers to the number of hidden units in each LSTM layer, with candidate values of 32, 64, and 128.These hyperparameters are critical in determining the forecasting performance of the model. However, increasing their values does not always yield improved accuracy, making it essential to identify an optimal parameter combination. Therefore, a grid search method (Fayed & Atiya, 2019) is employed to systematically explore the hyperparameter space. In this approach, predefined candidate values are assigned to each hyperparameter, all possible combinations are constructed, and the model is trained and evaluated under each configuration. The combination yielding the best predictive performance is selected as the optimal setting.

For clarity, the grid search results are grouped according to the number of LSTM hidden units: hidden_size = 32, 64, and 128. The corresponding results are reported in Tables 3, 4, and 5, respectively. As indicated by the experimental results, the configuration with seq_len = 20, horizon = 1, and hidden_size = 64 achieves the lowest forecasting error. Consequently, this parameter combination is adopted for the proposed GA-VMD-C-LSTM model and utilized in subsequent experiments to investigate the impact of the number of convolutional layers.

Table 3: Grid search results for LSTM with 32 hidden units.

Models	Hidden_Size	Seq_len	Horizon	MAE	RMSE	MAPE	R^2
GA-VMD-LSTM	32	5	1	24.7786	29.6090	0.5833	0.9972%
GA-VMD-LSTM	32	5	5	149.2741	179.2744	3.5026	0.8975%
GA-VMD-LSTM	32	5	10	120.1784	159.1873	2.8104	0.9194%
GA-VMD-LSTM	32	20	1	13.2543	17.4337	0.3075	0.9990%
GA-VMD-LSTM	32	20	5	72.1042	100.9550	1.7153	0.9677%
GA-VMD-LSTM	32	20	10	114.3935	158.2481	2.7015	0.9208%
GA-VMD-LSTM	32	60	1	68.4370	72.1879	1.6075	0.9837%
GA-VMD-LSTM	32	60	5	105.3982	136.0845	2.4820	0.9423%
GA-VMD-LSTM	32	60	10	103.1651	139.3278	2.4815	0.9397%

Table 4: Grid search results for LSTM with 64 hidden units.

Models	Hidden_Size	Seq_len	Horizon	MAE	RMSE	MAPE	R^2
GA-VMD-LSTM	64	5	1	16.8497	18.7256	0.4155	0.9989%
GA-VMD-LSTM	64	5	5	170.1812	190.6134	4.1865	0.8841%
GA-VMD-LSTM	64	5	10	100.2942	137.8637	2.4029	0.9396%
GA-VMD-LSTM	64	20	1	10.3502	12.9452	0.2490	0.9995%
GA-VMD-LSTM	64	20	5	83.8810	114.7721	1.9863	0.9582%
GA-VMD-LSTM	64	20	10	104.7047	145.5776	2.5170	0.9330%
GA-VMD-LSTM	64	60	1	33.3787	36.6420	0.7761	0.9958%
GA-VMD-LSTM	64	60	5	75.7746	106.7635	1.7969	0.9645%
GA-VMD-LSTM	64	60	10	109.0402	153.6357	2.6053	0.9267%

Table 5: Grid search results for LSTM with 128 hidden units.

Models	Hidden_Size	Seq_len	Horizon	MAE	RMSE	MAPE	R^2
GA-VMD-LSTM	128	5	1	61.1141	62.7204	1.5067	0.9874%
GA-VMD-LSTM	128	5	5	153.8862	178.7033	3.6691	0.8981%
GA-VMD-LSTM	128	5	10	163.2158	201.1092	3.8324	0.8714%
GA-VMD-LSTM	128	20	1	16.8130	19.1804	0.4288	0.9988%
GA-VMD-LSTM	128	20	5	79.3871	108.9263	1.8883	0.9624%
GA-VMD-LSTM	128	20	10	118.7126	155.9396	2.8267	0.9231%
GA-VMD-LSTM	128	60	1	42.9877	48.5961	0.9897	0.9926%
GA-VMD-LSTM	128	60	5	71.8329	101.0970	1.7161	0.9681%
GA-VMD-LSTM	128	60	10	108.1560	151.1375	2.5757	0.9290%

The second experiment investigates the impact of the number of convolutional layers on the forecasting performance of the proposed model. By comparing configurations with varying depths, we aim to demonstrate that integrating convolutional layers into the GA-VMD-LSTM architecture can enhance predictive accuracy and to identify the optimal convolutional depth for the hybrid model. Table 6 reports the forecasting results across different numbers of convolutional layers, and Fig. 8 illustrates the variation in evaluation metrics.

Table 6: Performance comparison of models with varying numbers of convolutional layers.

Models	MAE	RMSE	MAPE	R^2
GA-VMD-LSTM	10.3502	12.9452	0.2490%	0.9995
GA-VMD-C-LSTM(1)	5.5143	7.1673	0.1337%	0.9998
GA-VMD-C-LSTM(2)	7.0703	9.4908	0.1674%	0.9997
GA-VMD-C-LSTM(3)	96.1589	97.4192	2.3218%	0.9698
GA-VMD-C-LSTM(4)	227.1726	233.2729	5.4098%	0.8269
GA-VMD-C-LSTM(5)	32.2035	39.0334	0.7319%	0.9952
GA-VMD-C-LSTM(6)	85.9106	93.2007	2.1276%	0.9724

Note: The numbers in brackets in the first column indicate the number of convolutional layers used in the corresponding model.

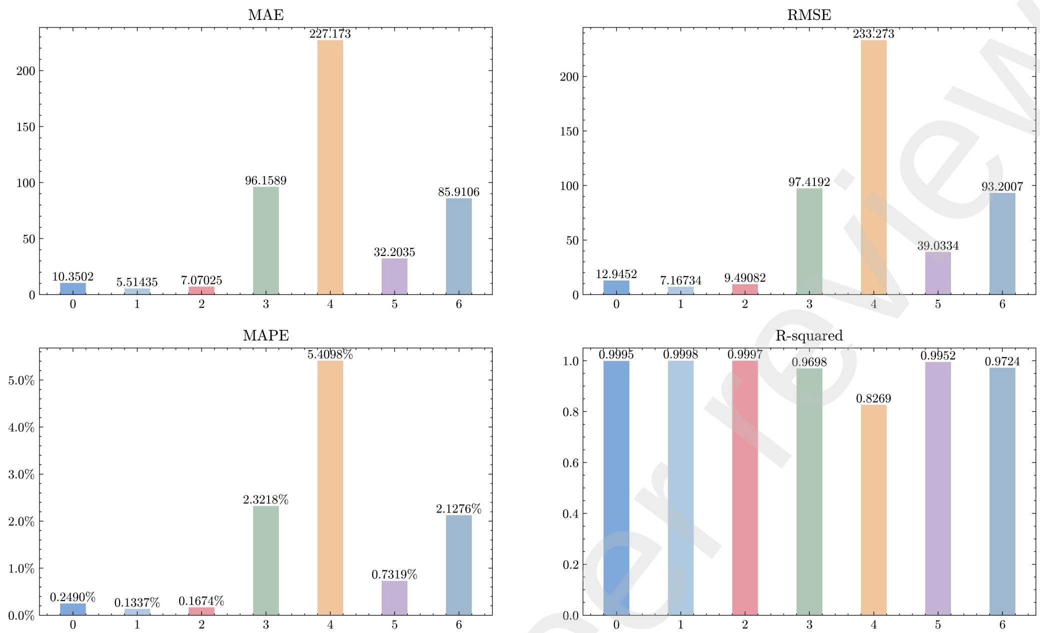

Figure 8: Trends in the number of convolutional layers and evaluation metrics.

As shown in Table 6, the GA-VMD-C-LSTM model with a single convolutional layer outperforms the other configurations. Observations from Fig. 8 indicate that, starting from zero layers, the MAE, RMSE, and MAPE generally increase while the R^2 value decreases as the number of convolutional layers grows. Interestingly, when the number of layers reaches five, the prediction errors decrease sharply; however, further increasing the depth to six layers leads to performance degradation. Although additional convolutional layers enable the model to capture more complex features, an excessive number of layers may adversely affect forecasting performance, potentially due to overfitting or increased model complexity.Overall, the results indicate that, compared with the baseline GA-VMD-LSTM model, the introduction of convolutional layers can enhance predictive performance. However, these layers should not be added arbitrarily. In our experiments, a single convolutional layer achieves the best performance and is therefore selected as the optimal configuration.

Building upon the previous findings regarding hyperparameter optimization and convolutional depth, the third experiment compares the forecasting errors of the proposed model with several benchmark models. Table 7 presents the MAE, RMSE, MAPE, and R^2 results for each model, while

Fig. 9 provides a visual comparison of their relative performance.

Table 7: Prediction error results for different models.

Models	MAE	RMSE	MAPE	R^2
BPNN	40.1888	57.5127	0.9596%	0.9895
MLP	45.5801	63.7348	1.0778%	0.9871
CNN	45.4285	65.0255	1.0868%	0.9866
LSTM	49.4923	68.0659	1.1938%	0.9853
GA-VMD-BPNN	27.3266	33.4796	0.6488%	0.9964
GA-VMD-MLP	46.5456	53.3333	1.1294%	0.9910
GA-VMD-CNN	19.2204	22.6964	0.4657%	0.9984
GA-VMD-LSTM	10.3502	12.9452	0.2490%	0.9995
GA-VMD-C-LSTM	5.5143	7.1673	0.1337%	0.9998

Figure 9: Prediction results from different models.

As shown in Table 7, with the exception of the GA-VMD-MLP model, which exhibits a slight increase in error compared to the standard MLP, all other benchmark models achieve a substantial reduction in prediction errors after incorporating the GA-optimized VMD module. This improvement is attributed to the ability of the VMD algorithm, when optimized by the genetic algorithm, to suppress noise and enhance the stationarity of stock price series. Moreover, the decomposed components reveal richer structural information, enabling subsequent models to learn more informative features and thereby improve predictive accuracy.Compared to the best-performing benchmark, GA-VMD-LSTM, the proposed GA-VMD-C-LSTM model achieves reductions of 46.72%, 44.63%, and 46.31% in MAE, RMSE, and MAPE, respectively, while the R^2 value increases by 0.03%. These results further confirm the effectiveness of integrating a convolutional layer into the GA-VMD-LSTM framework.

4.4. Trading decision application

To assess the practical applicability of the GA-VMD-C-LSTM model, a trading decision application based on the hybrid model was conducted. In the fourth experiment, a quantitative timing strategy was constructed as follows: each model was utilized to predict the closing price for the subsequent trading day, and execution signals were generated based on the sign and magnitude of the predicted return. If the predicted rate of change exceeded 0.001, indicating an expected price increase, the constituent stocks of the CSI 300 Index were purchased at the opening price of the following day. If the predicted rate of change was less than -0.001, indicating an expected price decrease, the holdings were liquidated at the opening price of the following day. If the predicted rate of change remained between -0.001 and 0.001, no action was taken.The strategy was implemented using the opening and closing prices of the CSI 300 Index, with initial capital set to 1 and transaction costs for both buying and selling fixed at 0.0005. The investment period spanned from August 9, 2019, to August 22, 2025, consistent with the test set utilized in the forecasting experiments. Furthermore, six sub-periods were selected for evaluation: August 2019–August 2020, August 2020–August 2021, August 2021–August 2022, August 2022–August 2023, August 2023–August 2024, and August 2024–August 2025. The timing strategy was back-tested using six performance indicators: initial net asset value, final net asset value, annualized return, overall annualized return, Sharpe ratio, and maximum drawdown. The evaluation results are reported in Tables 8 and 9, while a comparison of cumulative returns is illustrated in Fig. 10.

The annualized return represents the average annual growth rate of an investment over a given period, taking into account the effect of compounding. The calculation formula is shown in Eq. (24):

$$R_{\text{ann}} = \left(\frac{\text{NAV}_{\text{end}}}{\text{NAV}_{\text{start}}} \right)^{\frac{250}{N}} - 1 \tag{24}$$

Here, $\text{NAV}_{\text{start}}$ and NAV_{end} denote the net asset values at the beginning and end of each period, and N represents the actual number of trading days.

In accordance with market practice, one year is assumed to consist of 250 trading days.

The overall annualized return is computed based on the starting and ending net asset values of the concatenated evaluation periods and the total number of trading days, as shown in Eq. (25):

$$R_{\text{ann}}^{\text{total}} = \left(\frac{NAV_{\text{end}}^{\text{total}}}{NAV_{\text{start}}^{\text{total}}} \right)^{\frac{D_{\text{year}}}{D_{\text{total}}}} - 1 \tag{25}$$

Where $NAV_{\text{start}}^{\text{total}}$ and $NAV_{\text{end}}^{\text{total}}$ denote the net asset values at the first and last trading days of the entire evaluation horizon, respectively; D_{total} is the total number of trading days across all evaluation periods; and D_{year} represents the number of trading days in one year.

The Sharpe ratio is employed to measure the risk-adjusted return of the strategy (Sharpe, 1966).The calculation formula is shown in Eq. (26):

$$\text{Sharpe} = \frac{\mathbb{E}(R) - R_f}{\sigma(R)} \tag{26}$$

where $\mathbb{E}(R)$ denotes the expected return of the strategy, calculated as the annualized mean of daily simple returns; R_f is the annualized risk-free rate; and $\sigma(R)$ represents the annualized standard deviation of returns.

The maximum drawdown is defined as the maximum peak-to-trough decline in the net asset value over the entire evaluation period (Grossman & Zhou, 1993). The calculation formula is shown in Eq. (27):

$$\text{MDD} = \min_t \left(\frac{NAV_t - \max_{0 \le i \le t} NAV_i}{\max_{0 \le i \le t} NAV_i} \right) \tag{27}$$

where NAV_t denotes the net asset value at time t, and $\max_{0 \le i \le t} NAV_i$ is the historical maximum net asset value up to time t.

Table 8: Performance evaluation metrics for different timing strategies across intervals.

Model	Period	Start Date	End Date	Initial Net Value	Final Net Value	Annualized Return
Buy-and-Hold Strategy	2019-08-22 to 2020-08-22	2019-08-22	2020-08-24	1.0390	1.2999	25.80%
Buy-and-Hold Strategy	2020-08-22 to 2021-08-22	2020-08-24	2021-08-23	1.2999	1.3085	0.68%
Buy-and-Hold Strategy	2021-08-22 to 2022-08-22	2021-08-23	2022-08-22	1.3085	1.1325	-13.86%
Buy-and-Hold Strategy	2022-08-22 to 2023-08-22	2022-08-22	2023-08-22	1.1325	1.0265	-9.58%
Buy-and-Hold Strategy	2023-08-22 to 2024-08-22	2023-08-22	2024-08-22	1.0265	0.9100	-11.61%
Buy-and-Hold Strategy	2024-08-22 to 2025-08-22	2024-08-22	2025-08-22	0.9100	1.1742	29.98%
Buy-and-Hold Strategy	2019-08-22 to 2025-08-22	2019-08-22	2025-08-22	1.0390	1.1742	2.06%
GA-VMD-C-LSTM	2019-08-22 to 2020-08-22	2019-08-22	2020-08-24	0.9880	1.6956	73.91%
GA-VMD-C-LSTM	2020-08-22 to 2021-08-22	2020-08-24	2021-08-23	1.6956	1.6748	-1.25%
GA-VMD-C-LSTM	2021-08-22 to 2022-08-22	2021-08-23	2022-08-22	1.6748	1.6164	-3.60%
GA-VMD-C-LSTM	2022-08-22 to 2023-08-22	2022-08-22	2023-08-22	1.6164	3.2391	103.85%
GA-VMD-C-LSTM	2023-08-22 to 2024-08-22	2023-08-22	2024-08-22	3.2391	3.3603	3.84%
GA-VMD-C-LSTM	2024-08-22 to 2025-08-22	2024-08-22	2025-08-22	3.3603	4.5644	37.04%
GA-VMD-C-LSTM	2019-08-22 to 2025-08-22	2019-08-22	2025-08-22	0.9880	4.5644	29.05%
GA-VMD-LSTM	2019-08-22 to 2020-08-22	2019-08-22	2020-08-24	0.9758	1.5506	60.73%
GA-VMD-LSTM	2020-08-22 to 2021-08-22	2020-08-24	2021-08-23	1.5506	1.5533	0.18%
GA-VMD-LSTM	2021-08-22 to 2022-08-22	2021-08-23	2022-08-22	1.5533	1.3599	-12.84%
GA-VMD-LSTM	2022-08-22 to 2023-08-22	2022-08-22	2023-08-22	1.3599	2.5406	89.72%
GA-VMD-LSTM	2023-08-22 to 2024-08-22	2023-08-22	2024-08-22	2.5406	2.4859	-2.20%
GA-VMD-LSTM	2024-08-22 to 2025-08-22	2024-08-22	2025-08-22	2.4859	3.3295	35.07%
GA-VMD-LSTM	2019-08-22 to 2025-08-22	2019-08-22	2025-08-22	0.9758	3.3295	22.70%
GA-VMD-BPNN	2019-08-22 to 2020-08-22	2019-08-22	2020-08-24	1.0147	1.0843	13.79%
GA-VMD-BPNN	2020-08-22 to 2021-08-22	2020-08-24	2021-08-23	1.0843	1.1643	7.57%
GA-VMD-BPNN	2021-08-22 to 2022-08-22	2021-08-23	2022-08-22	1.1643	1.1036	-5.38%
GA-VMD-BPNN	2022-08-22 to 2023-08-22	2022-08-22	2023-08-22	1.1036	1.9680	80.88%
GA-VMD-BPNN	2023-08-22 to 2024-08-22	2023-08-22	2024-08-22	1.9680	2.1879	11.46%
GA-VMD-BPNN	2024-08-22 to 2025-08-22	2024-08-22	2025-08-22	2.1879	2.6436	21.49%
GA-VMD-BPNN	2019-08-22 to 2025-08-22	2019-08-22	2025-08-22	1.0843	2.6436	18.48%
GA-VMD-CNN	2019-08-22 to 2020-08-22	2019-08-22	2020-08-24	0.9723	1.5507	61.33%
GA-VMD-CNN	2020-08-22 to 2021-08-22	2020-08-24	2021-08-23	1.5507	1.4393	-7.35%
GA-VMD-CNN	2021-08-22 to 2022-08-22	2021-08-23	2022-08-22	1.4393	1.9888	39.66%
GA-VMD-CNN	2022-08-22 to 2023-08-22	2022-08-22	2023-08-22	1.9888	3.0087	52.83%
GA-VMD-CNN	2023-08-22 to 2024-08-22	2023-08-22	2024-08-22	3.0087	2.9475	-2.08%
GA-VMD-CNN	2024-08-22 to 2025-08-22	2024-08-22	2025-08-22	2.9475	3.9895	36.53%
GA-VMD-CNN	2019-08-22 to 2025-08-22	2019-08-22	2025-08-22	0.9723	3.9895	26.53%
GA-VMD-MLP	2019-08-22 to 2020-08-22	2019-08-22	2020-08-24	0.9559	1.0025	5.00%
GA-VMD-MLP	2020-08-22 to 2021-08-22	2020-08-24	2021-08-23	1.0025	1.1791	18.08%
GA-VMD-MLP	2021-08-22 to 2022-08-22	2021-08-23	2022-08-22	1.1791	1.0382	-12.32%
GA-VMD-MLP	2022-08-22 to 2023-08-22	2022-08-22	2023-08-22	1.0382	1.5965	55.42%
GA-VMD-MLP	2023-08-22 to 2024-08-22	2023-08-22	2024-08-22	1.5965	1.9517	22.85%
GA-VMD-MLP	2024-08-22 to 2025-08-22	2024-08-22	2025-08-22	1.9517	2.7125	40.30%
GA-VMD-MLP	2019-08-22 to 2025-08-22	2019-08-22	2025-08-22	0.9559	2.7125	18.99%

Table 9: Comprehensive evaluation metrics for timing strategies across different models.

Model	Period	Initial Net Value	Final Net Value	Overall Annualized Return	SR	MDD (%)
Buy-and-Hold Strategy	2019-08-22 to 2025-08-22	1.0390	1.1742	2.11%	0.2042	-46.66%
GA-VMD-C-LSTM	2019-08-22 to 2025-08-22	0.9880	4.5644	29.94%	1.4072	-23.03%
GA-VMD-LSTM	2019-08-22 to 2025-08-22	0.9758	3.3295	23.37%	1.1336	-21.43%
GA-VMD-BPNN	2019-08-22 to 2025-08-22	1.0843	2.6436	19.01%	0.9567	-22.27%
GA-VMD-CNN	2019-08-22 to 2025-08-22	0.9723	3.9895	27.33%	1.3057	-23.06%
GA-VMD-MLP	2019-08-22 to 2025-08-22	0.9559	2.7125	19.54%	0.9853	-21.77%

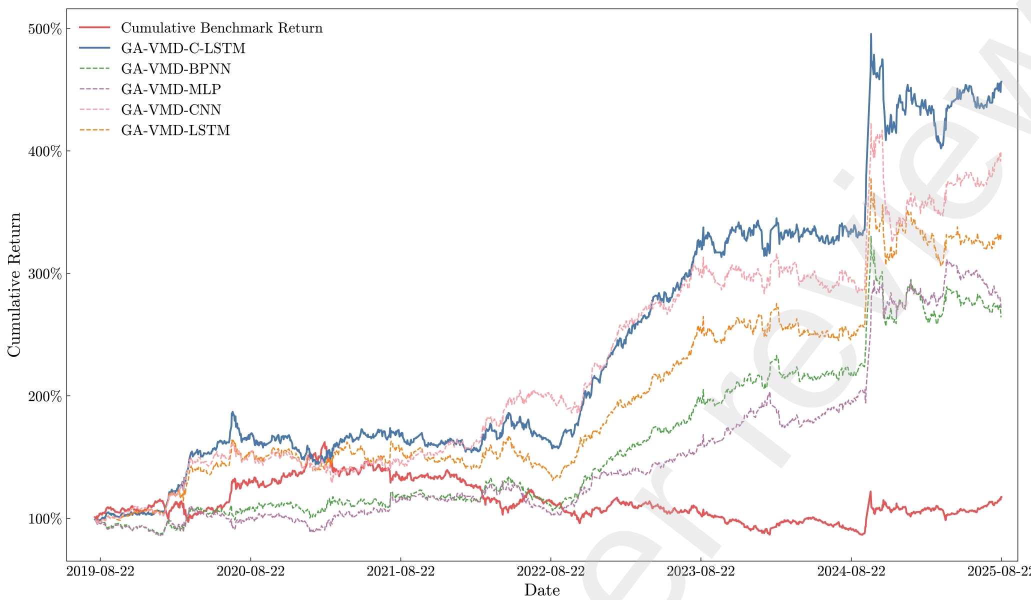

Figure 10: Cumulative returns of timing strategies across different models.

Based on Table 8, it can be observed that the GA-VMD-C-LSTM model achieves a Sharpe ratio of 1.4072, which is higher than those of all benchmark models. The proposed model attains a final net asset value of 4.5644 on August 22, 2025, a figure that is higher than that of the buy-and-hold strategy (1.1742) and exceeds the final net asset values of the other benchmark models. In addition, the GA-VMD-C-LSTM model achieves the highest annualized return of 103.85%, with an overall annualized return of 29.94%, compared to 2.11% for the buy-and-hold strategy and lower values for the remaining benchmark models. This performance is attributed to the proposed model's capability to predict next-day closing prices with greater accuracy than the benchmark models.While the maximum drawdown of the GA-VMD-C-LSTM model is relatively higher than that of certain benchmark models, this may be related to the generally higher daily strategy net values, which can lead to larger fluctuations in absolute terms. As shown in Fig. 10, the cumulative return of the GA-VMD-C-LSTM model remains above those of the buy-and-hold strategy and the other benchmark models for the majority of the sample period. Specifically, the cumulative return of the proposed model reaches 456.44% on August 22, 2025, compared to 117.42% for the buy-and-hold strategy and lower returns for the benchmark models. The results of the timing strategy experiment indicate that the proposed GA-VMD-C-

LSTM forecasting model possesses potential for application in the field of quantitative investment.

5. Conclusion

As a critical component of capital markets, the stock market is intricately linked to socioeconomic development, making price forecasting essential for asset investment and risk management. Based on a review of existing literature, several limitations are identified.Firstly, correlations among the components obtained via Variational Mode Decomposition (VMD) are often neglected.Secondly,VMD parameter selection based on empirical rules or trial-and-error lacks guaranteed decomposition quality.Thirdly,standard LSTM networks possess limited capacity for extracting local features from time series data.Fourthly, existing research on machine learning models for stock price forecasting largely neglects to explore the practical application value of these models.

To address these gaps, this study introduces the following innovations:(1) A genetic algorithm (GA) is employed to search for optimal VMD parameters (K, α, τ), thereby enhancing decomposition quality.(2) VMD components generated with optimized parameters are jointly fed into the neural network, rather than being modeled separately, enabling the exploitation of cross-component correlations.(3) Convolutional layers are incorporated into the neural network architecture to extract local features, and their impact on forecasting performance is systematically analyzed.(4) A market timing strategy is implemented to evaluate the returns of the proposed model in a simulated trading environment, thereby validating its practical applicability and robustness.

In addition, this study proposes a trading decision system for stock index closing price forecasting and timing. The decision support system consists of two main components. The first component is a hybrid stock price forecasting model that integrates a genetic algorithm, variational mode decomposition (VMD), and neural networks. The model architecture is divided into two stages. In the data preprocessing stage, a genetic algorithm is employed to search for the optimal VMD parameters, and the optimized VMD is then used to decompose the closing price series into multiple intrinsic mode functions (IMFs) to reveal latent feature information. In the neural network stage, convolutional layers in conjunction with LSTM layers are utilized to extract local and sequential information from the time series to perform forecasting, thereby improving prediction performance.The second component focuses on trading decision applications based on the proposed model. Specifically, the GA-VMD-C-LSTM model is used to predict the next-day closing price and generate trading signals, and excess returns are targeted by executing buy and sell decisions at identified intervals.

A series of experiments was conducted using the CSI 300 Index closing prices, yielding the following results:(1) Grid search was applied to determine the optimal values of seq_len, horizon, and hidden_size for the baseline GA-VMD-LSTM model. The empirical results indicate that seq_len = 20, horizon = 1, and hidden_size = 64 constitute the optimal parameter combination with the lowest prediction error.(2) Based on the optimal configuration obtained in the first experiment, different numbers of convolutional layers (1–6) were introduced into the GA-VMD-LSTM framework to construct multiple GA-VMD-C-LSTM variants. The results show that the GA-VMD-C-LSTM model with a single convolutional layer achieves the lowest forecasting error and the best overall performance, confirming the effectiveness of incorporating convolutional layers. This configuration is therefore adopted as the proposed model.(3) The proposed GA-VMD-C-LSTM model was compared with eight benchmark models, including BPNN, MLP, CNN, LSTM, GA-VMD-BPNN, GA-VMD-MLP, GA-VMD-CNN, and GA-VMD-LSTM. The experimental results demonstrate that the GA-VMD-C-LSTM model with one convolutional layer achieves the best forecasting performance. Compared with the best-performing benchmark, GA-VMD-LSTM, the MAE, RMSE, and MAPE are reduced by 46.72%, 44.63%, and 46.31%, respectively, while the R^2 value increases by 0.03%.(4) To assess the practical application value of the proposed model, a trading decision experiment was conducted. Specifically, a quantitative timing strategy was constructed, and five models, including GA-VMD-BPNN, GA-VMD-MLP, GA-VMD-CNN, GA-VMD-LSTM, and GA-VMD-C-LSTM, were selected for a comparative evaluation of their returns. The results indicate that the proposed GA-VMD-C-LSTM model achieves the highest cumulative return on the test set, reaching 456.44%, demonstrating its potential for quantitative investment practice.

Stock price forecasting is widely regarded as a challenging yet essential task for investors. The proposed decision support system provides a forecasting model that integrates optimization algorithms, frequency decomposition techniques, and neural networks, thereby offering an alternative approach to stock price prediction. Furthermore, a quantitative timing strategy is developed based on this model, enabling the framework to be applied in practice. Investors can utilize the proposed decision support system in quantitative investment, where the forecasting results may assist in making buy and sell decisions, potentially mitigating investment risk and achieving excess returns.

Although the proposed decision support system has achieved the recorded performance, several avenues remain for extending this work. In future research, additional features, such as political and macroeconomic indicators, could be incorporated into the forecasting model to improve its adaptability under extreme market conditions. While the proposed hybrid model is designed to forecast stock index closing prices, it can be extended to predict the prices of other financial instruments, such as options, futures, and bonds. Regarding trading decision applications, further refinements could be achieved by incorporating transaction-related costs (e.g., stamp duty), implementing take-profit and stop-loss mechanisms, and introducing dynamic position adjustment schemes into the quantitative timing strategy to optimize overall strategy performance.

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